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MACHINE LEARNING ASSIGNMENT REPORT

**HIGGS Boson Classification Using
Supervised Machine Learning Techniques**

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HIGGS Boson Classification Using Supervised Machine Learning Techniques

ABSTRACT

Machine Learning has emerged as a powerful technology for solving complex problems in scientific research, data analysis, and predictive modelling. One of its most important applications is in particle physics, where massive volumes of experimental data must be analysed to identify rare particle events. The Higgs Boson, a fundamental particle in the Standard Model of Physics, plays a crucial role in understanding the origin of mass. Detecting Higgs Boson events is challenging because they are hidden among a large number of background events.

This project focuses on the classification of Higgs Boson events using supervised machine learning techniques. The HIGGS Dataset, obtained from the UCI Machine Learning Repository, is used for the study. The project begins with comprehensive Exploratory Data Analysis (EDA), including data understanding, preprocessing, visualization, feature analysis, and correlation studies to gain insights into the dataset.

Three supervised classification algorithms—Logistic Regression, Random Forest Classifier, and Support Vector Machine (SVM)—are implemented and evaluated. Model performance is assessed using metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and Classification Report. A comparative analysis is conducted to identify the most effective model.

The results demonstrate that machine learning algorithms can successfully distinguish Higgs Boson signal events from background events, highlighting the significance of machine learning in addressing large-scale scientific classification challenges.

1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without explicit programming. Machine learning techniques are widely used in various fields such as healthcare, finance, robotics, image processing, and scientific research.

The discovery of the Higgs Boson particle is one of the most important achievements in modern particle physics. Identifying Higgs Boson events from millions of particle collision records is a challenging task due to the presence of a large number of background events. Traditional analysis methods are often time-consuming and may not efficiently handle large-scale datasets. Therefore, machine learning techniques are increasingly being used to improve classification accuracy and data analysis in particle physics.

This project applies supervised machine learning techniques to classify Higgs Boson events using features available in the HIGGS dataset. The project follows a complete machine learning workflow including:

- Data Collection
- Exploratory Data Analysis (EDA)
- Data Preprocessing
- Feature Analysis
- Model Training
- Model Evaluation
- Comparative Performance Analysis

The objective of this project is to develop and compare different supervised machine learning models for accurate classification of Higgs Boson signal events and background events, thereby demonstrating the effectiveness of machine learning in scientific data analysis.

DATASET DESCRIPTION

The HIGGS Dataset is a large-scale machine learning dataset developed for the classification of particle collision events in high-energy physics experiments. The dataset is obtained from the UCI Machine Learning Repository and is widely used for evaluating classification algorithms on large and complex scientific data.

The primary objective of the dataset is to distinguish between events that contain evidence of the Higgs Boson particle and events that do not. This is formulated as a binary classification problem, where each event is assigned one of two class labels. A class label of **1** represents a Higgs Boson signal event, indicating that the collision may have produced a Higgs Boson particle. A class label of **0** represents a background event, meaning that the observed event is generated by other physical processes and does not contain a Higgs Boson signal.

The dataset contains approximately **11 million instances**, making it one of the largest publicly available datasets for machine learning research. Each instance consists of **28 continuous numerical features** and one target class label. The first **21 features** are low-level kinematic properties directly measured by particle detectors during collision experiments. These measurements include information such as particle momentum, energy, and direction. The remaining **7 features** are high-level features derived by physicists using combinations of the low-level measurements to improve the distinction between signal and background events.

Due to its large size and scientific importance, the HIGGS dataset serves as an excellent benchmark for evaluating machine learning algorithms and demonstrates the practical application of artificial intelligence in particle physics research.

PROBLEM STATEMENT

The discovery of the Higgs Boson is one of the most significant achievements in modern particle physics. However, identifying Higgs Boson events from particle collision data is a challenging task because these events are extremely rare and are often hidden among a large number of background events. High-energy physics experiments generate millions of collision records, making manual analysis impractical and time-consuming.

The HIGGS dataset presents this challenge as a binary classification problem. Each collision event is represented by 28 numerical features describing various physical properties measured during the experiment. The objective is to determine whether a given event corresponds to a Higgs Boson signal or a background event.

In this project, supervised machine learning techniques are employed to automatically classify particle collision events into two categories:

- **Class 1 (Signal Event):** Indicates the presence of a Higgs Boson event.
- **Class 0 (Background Event):** Indicates the absence of a Higgs Boson event.

The primary goal of this study is to develop accurate and reliable classification models capable of distinguishing signal events from background events. The performance of the models is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. By minimizing classification errors and maximizing prediction accuracy, the project demonstrates the effectiveness of machine learning in solving large-scale scientific classification problems.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is an important step in the machine learning workflow that helps in understanding the structure, quality, and characteristics of the dataset before model development. The primary objective of EDA is to identify patterns, relationships, distributions, and potential issues within the data that may affect model performance.

For the HIGGS dataset, EDA was performed to gain insights into the particle collision data and understand how the features contribute to the classification of Higgs Boson signal events and background events. The analysis began with **dataset inspection**, where the number of records, features, data types, and missing values were examined. This helped verify that the dataset contained 11 million instances with 28 continuous numerical features and one binary target variable.

Next, **class distribution analysis** was conducted to study the balance between signal events (Class 1) and background events (Class 0). Understanding class distribution is essential because highly imbalanced datasets can lead to biased model predictions.

Histograms were plotted to visualize the distribution of feature values and identify skewness, spread, and variations among different features. **Correlation analysis** was performed using a heatmap to examine relationships between features and detect any strong dependencies. **Boxplots** were used to identify outliers and understand feature variability.

The EDA process provided valuable insights into the dataset, helping guide preprocessing, feature scaling, model selection, and overall machine learning strategy.

EDA OBSERVATIONS

The Exploratory Data Analysis (EDA) of the HIGGS dataset provided important insights into the characteristics of the data and helped in understanding the relationships between features before model training.

Class Distribution Analysis

The class distribution plot showed that both classes are relatively balanced, with a comparable number of signal events (Class 1) and background events (Class 0). This indicates that the dataset does not suffer from severe class imbalance. As a result, machine learning models can learn patterns from both classes effectively without requiring additional balancing techniques such as oversampling or under sampling.

Histogram Analysis

Histograms were generated for multiple features to study their distributions. The analysis revealed that most features do not follow a perfect normal distribution. Several features exhibit positive or negative skewness, indicating asymmetrical distributions. The spread of values varies significantly across different features, suggesting that some features operate on larger numerical scales than others. These observations justify the use of feature scaling before training models such as Logistic Regression and Linear SVM.

Boxplot Analysis

Boxplots were used to identify outliers and understand feature variability. A considerable number of outliers were observed across several features. In particle physics datasets, such extreme values are expected because particle collision events can vary greatly in their physical characteristics. Therefore, these outliers represent meaningful scientific observations rather than data entry errors and were retained for model training.

Overall Observation

The EDA results indicate that the dataset is suitable for machine learning classification. The balanced class distribution, diverse feature patterns, and presence of meaningful variations provide valuable information for distinguishing Higgs Boson signal events from background events.

CORRELATION HEATMAP ANALYSIS

A correlation heatmap was generated to examine the relationships among the features in the HIGGS dataset. Correlation analysis helps identify how strongly different variables are related to one another and whether any features provide redundant information. Understanding these relationships is important because highly correlated features can sometimes lead to multicollinearity issues, which may negatively affect the performance of certain machine learning models.

The heatmap revealed that most feature pairs exhibit **weak to moderate correlations**, indicating that the features capture different aspects of particle collision events. Only a few feature combinations showed relatively stronger correlations, while the majority remained independent or only mildly related. This suggests that the dataset contains diverse information that can contribute to the classification process.

Another important observation is that **severe multicollinearity was not observed** among the examined features. Since no large groups of features were found to be strongly correlated with each other, there was no immediate need for feature removal based on correlation alone. Each feature appears to provide unique information that may help distinguish Higgs Boson signal events from background events.

The correlation analysis also indicates that the classification problem is complex and cannot be solved using only a few variables. Instead, multiple features collectively contribute to identifying patterns associated with Higgs Boson events.

Conclusion

The correlation heatmap demonstrates that the HIGGS dataset contains rich and diverse feature information. Most features contribute independently to the prediction task, making them valuable inputs for machine learning models such as Logistic Regression, Random Forest, and Linear SVM. This diversity of information helps improve the ability of the models to accurately classify signal and background events.

DATA PREPROCESSING

Data preprocessing is a crucial step in machine learning because raw data cannot be directly used for model training. Proper preprocessing improves data quality, ensures consistency, and enhances model performance. In this project, several preprocessing steps were performed before applying machine learning algorithms to the HIGGS dataset.

The first step involved **handling the dataset structure**. The dataset was loaded into a Pandas DataFrame, and appropriate column names were assigned. The first column represents the target class label, while the remaining 28 columns represent the input features describing particle collision events.

Next, **feature-target separation** was performed. The target variable (class label) was stored separately as the dependent variable (y), while all feature columns were stored as independent variables (X). This separation is necessary because machine learning models learn patterns from input features to predict the target class.

After separating features and labels, the dataset was divided into **training and testing sets** using a train-test split. The training set was used to train the models, while the testing set was reserved for evaluating model performance on unseen data. This ensures that the models are assessed fairly and helps prevent overfitting.

The final preprocessing step was **feature scaling using StandardScaler**. Since different features have different numerical ranges, scaling standardizes the data by transforming each feature to have a mean of zero and a standard deviation of one. Feature scaling is particularly important for **Logistic Regression** and **Linear Support Vector Machine (SVM)** because these algorithms are sensitive to feature magnitudes. Scaling improves convergence speed, numerical stability, and overall model accuracy.

Thus, preprocessing ensured that the HIGGS dataset was properly prepared for efficient and reliable machine learning model training.

SAMPLING STRATEGY AND TRAIN-TEST SPLIT

The HIGGS dataset contains approximately **11 million particle collision events**, making it one of the largest publicly available datasets for machine learning research. Although the large dataset size provides extensive information for model training, processing all 11 million records requires significant computational resources, memory, and training time. Since this project was performed on a standard computing environment, a sampling strategy was adopted to make the analysis computationally feasible.

A representative subset of the original dataset was selected while preserving the overall characteristics and class distribution of the data. Using a subset significantly reduced memory consumption and training time without affecting the ability to evaluate machine learning models effectively. This approach is commonly used in large-scale machine learning projects when hardware resources are limited.

After sampling, the dataset was divided into **training and testing sets using an 80:20 ratio**. The training set consisted of 80% of the selected records and was used for learning patterns and relationships within the data. The remaining 20% of the data was reserved as the testing set and was used exclusively for evaluating model performance on unseen data.

The use of a train-test split provides a fair assessment of model generalization and helps detect overfitting. An 80:20 split is widely adopted in machine learning because it provides sufficient training data while maintaining a large enough testing set for reliable evaluation.

Reasons for Sampling and Train-Test Splitting

- Faster model training and experimentation.
- Reduced memory and computational requirements.
- Fair and unbiased model evaluation.
- Better assessment of model performance on unseen data.
- Prevention of overfitting and improved generalization.

MACHINE LEARNING MODELS USED

Three supervised machine learning algorithms were selected for this project to classify particle collision events as Higgs Boson signal events or background events. Each model was chosen based on its unique characteristics and suitability for binary classification problems.

1. Logistic Regression

Logistic Regression is a widely used classification algorithm that predicts the probability of an event belonging to a particular class. It serves as a simple and interpretable baseline model. The algorithm is computationally efficient and performs well when relationships between features and target classes are approximately linear. It provides a useful benchmark for comparing more advanced machine learning models.

2. Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance. It can capture complex nonlinear relationships and interactions among features. Random Forest is robust to noise and overfitting and often provides high classification accuracy on large datasets. It also offers feature importance information, which helps identify influential variables.

3. Linear Support Vector Machine (SVM)

Linear SVM is a powerful classification algorithm that finds the optimal decision boundary separating classes. It is particularly effective for high-dimensional datasets and can handle large numbers of features efficiently. SVM aims to maximize the margin between classes, improving generalization on unseen data.

These three models were selected to compare linear and nonlinear classification approaches and determine the most effective technique for Higgs Boson event classification.

ENSEMBLE METHODS AND CLASSIFICATION METRICS

Ensemble Methods

Ensemble learning is a machine learning technique that combines multiple models to improve prediction accuracy and reliability. Instead of relying on a single model, ensemble methods aggregate predictions from several models to produce a stronger final prediction.

Random Forest is an ensemble method based on Decision Trees. It uses a technique called Bootstrap Aggregation (Bagging), where multiple random samples are drawn from the training data. A separate decision tree is trained on each sample. During prediction, every tree votes for a class label, and the class receiving the majority of votes becomes the final prediction.

Advantages of Random Forest include:

- Higher classification accuracy.
- Better generalization to unseen data.
- Reduced risk of overfitting.
- Ability to handle complex nonlinear relationships.

Classification Metrics

Accuracy measures the proportion of correctly classified instances among all predictions.

Precision measures the proportion of correctly predicted positive cases among all positive predictions.

Recall measures the ability of a model to correctly identify actual positive instances.

F1-Score is the harmonic mean of Precision and Recall and provides a balanced measure of performance.

Support indicates the number of actual observations belonging to each class.

Macro Average calculates the average metric value across classes without considering class frequency.

Weighted Average calculates the average metric value while considering class frequency.

The Confusion Matrix summarizes classification performance using:

- True Positives (TP)
- True Negatives (TN)

- False Positives (FP)
- False Negatives (FN)

These metrics provide a comprehensive evaluation of model performance beyond simple accuracy.

MODEL RESULTS, COMPARISON, CONCLUSION AND REFERENCES

Model Results and Comparison

The performance of Logistic Regression, Random Forest, and Linear SVM was evaluated using Accuracy, Precision, Recall, F1-Score, Classification Reports, and Confusion Matrices.

The confusion matrices show how effectively each model classified signal and background events. Logistic Regression provided a strong baseline performance and demonstrated good computational efficiency. Linear SVM achieved competitive classification results and effectively handled the high-dimensional feature space. Random Forest achieved the highest overall performance by capturing nonlinear relationships and complex interactions among features.

Comparative analysis indicates that Random Forest consistently produced superior classification results, making it the most suitable model for the HIGGS dataset.

Conclusion

This project successfully applied machine learning techniques to classify Higgs Boson signal events and background events using the HIGGS dataset. Exploratory Data Analysis helped understand feature distributions, correlations, and data characteristics. Feature scaling improved the performance of Logistic Regression and Linear SVM. Among the three evaluated models, Random Forest demonstrated the best classification performance due to its ensemble learning capability and ability to model complex patterns.

The study highlights the importance of machine learning in particle physics and demonstrates how data-driven methods can assist in large-scale scientific discoveries.

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